



# Proposed CEA Link — December 2026

A proposal for the CEA link on the December 2026 timetable, judged against today's link and the December 2026 rules

Prepared 28 September 2026 for the December 2026 timetable · assessed by the Links designer's own rule modules · every figure is computed from the grid, not typed

## PROPOSAL

# Same Turns

Today's link widened to 24 lines, in today's own shift times

CODE **ST-24-B7**

FINGERPRINT **d15e1b74**

BUILT FROM 24 lines · 4 cover weeks · computer search

✓ **24** lines designed

✓ **35h 00m** a week — exactly the contract

✓ **0** hard-limit breaches

✓ **0** rests under 12h

⚠ **1** fatigue factor present (today: 5)

Five questions a manager asks first, in order. Every figure is computed from the cells; **page 2 says how to read each one**, with today's link and the other proposals marked on a scale.

### CAN IT BE RUN?

## Yes

the hard limits

0 rests under 12h · longest run 6 of 13 · contract exact

### DOES IT MEET THE DECEMBER SHAPE?

## 7 of 11

headcount rules met

the staffing levels, the ticket office, the floor and the handovers — page 7 · today's link meets 3 of 11

### HOW TIRING IS IT?

## 1

fatigue factors present, of 25

advisory — present means worth a look, not a breach · today 5 · fewest of the 23 0

### DOES IT FOLLOW THE TRAINS?

## 46.3

weekday demand fit — lower is closer

Saturday 32.7 · Sunday 80.6 · today 51.1 · best of the 23 21.9 · on the floor 57.5 (today 68.8)

### IS IT FAMILIAR?

## 15 of 15

shift times people work today

nothing new to learn · 18 of 20 weeks are one turn (today 7 of 16)

## 6 in 24

full weekends off

25% of the rotation, against 20% today (4 in 20) · most of the 23 25%

## 4 cover weeks

lines 1, 7, 13, 19

evenly spaced — gaps of 6, 6, 6 lines · today's sit at 1, 7, 12, 17

## 2h 33m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

## What this is

**Same Turns** is a proposal for the CEA link on the December 2026 timetable: a 24-line rotation for 24 people, where today's link has 20. Today's link widened to 24 lines, in today's own shift times. Built by computer search: the duty table was chosen for how closely it follows the December 2026 timetable, and the 24 weeks put in the order that raises the fewest fatigue factors.

**Against today's link.** It meets **7 of the 11** December 2026 rules, where today's link meets 3; it has 14 on a Saturday and 10 on a Sunday (today 10 and 8). With the ticket office taken out, the people on the floor follow the weekday trains more closely than today's (a fit of 57.5 against 68.8; lower is closer). 1 ORR fatigue factor is present, against 5 today. It works 15 shift times, 15 of them worked today.

**What it is not.** Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December 2026 headcounts are relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 7 and 8 say so where it applies. It clears every hard limit the tool can check — twelve hours between duties, the 13-day limit and the exact contract — which is the floor for being worth discussing.

## HOW TO READ THE NUMBERS

# What each figure means, and what good looks like

Every figure on this sheet is computed from the cells. On each scale: this sheet in navy, today's 20-line link as a ring, the best and worst of the other 22 proposals as gold and grey ticks.

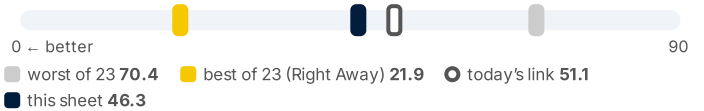
**Three things to hold.** Lower is better on the fit and the fatigue count; higher is better on rest, weekends, one-turn weeks and familiarity. "Present" is not "failed" — the ORR list reports what is there. Today's link is the yardstick, not the target: it meets none of the December 2026 headcounts.

### DOES IT FOLLOW THE TRAINS?

#### Demand fit — weekday

How closely the people on duty follow the trains through an average weekday. For each of its 18 hours, the hour's share of the staff-minutes is set against its share of the traffic; the gaps, in percentage points, are squared and added up — 0 would be a perfect match. The typical hourly gap is the square root of the score ÷ 18: about 1.6 points here, 1.7 today.

**What it does not tell you.** It compares shares, not headcount — adding people evenly to every hour changes nothing — and the station rosters posts that need someone whether the hour carries 23 trains or 5, so a flatter day can be right and still score worse.



Saturday 32.7 (today 45.7, best of 23 8.1) · Sunday 80.6 (today 65.9, best 20.9). Each day type has its own curve and window, so a Saturday is only compared with a Saturday. **Floor**, ticket office out of both: 57.5 · 48.7 · 100.9 (today 68.8 · 68.7 · 72.5).

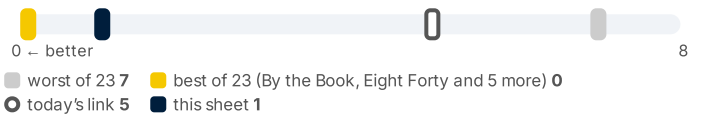
On page 6.

### HOW TIRING IS IT?

#### Fatigue factors present

How many of the 25 roster patterns on the Office of Rail and Road's good-practice list appear anywhere in the rotation — long runs of earlys, short gaps between duties, start times that jump — counted by the Links designer's own module.

**What it does not tell you.** A factor present is worth a look, never a breach: the ORR says the list is guidance, not limits. Zero does not mean approved and one does not mean unsafe; page 8 names which, and how big.



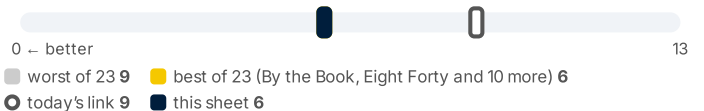
On page 8.

### CAN IT BE RUN?

#### Longest run of worked days

The most consecutive days anyone works in the worst case — a cover week's four duties placed as badly as they can be. Chiltern's limit is 13; the Links designer's own target is 6.

**What it does not tell you.** The roster clerk decides where cover-week duties fall, so the as-rostered figure can be lower; page 8 shows both readings.



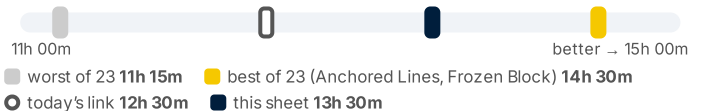
On page 7.

### CAN IT BE RUN?

#### Tightest rest between duties

The shortest gap anywhere between one duty ending and the next beginning — Saturday into Sunday and the last line into the first included. Twelve hours is the floor; a design under it cannot be run.

**What it does not tell you.** It is one gap. A rotation with many 12h05 rests and one with none read the same here; the count under 12h is in the chips on page 1.



On page 7.

### WHAT IS IT LIKE TO WORK?

#### Full weekends off

The share of weeks with both Saturday and Sunday off. Today's link gives 4 in 20, one weekend in 5; this sheet gives 6 in 24, one weekend in 4.

**What it does not tell you.** A weekend off beside a 42-hour week is still a weekend off; the hours by week are on the grid, page 4.



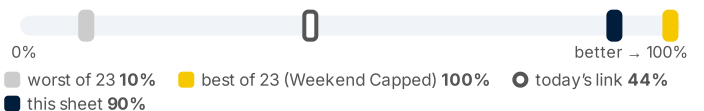
On page 5.

### WHAT IS IT LIKE TO WORK?

#### Working weeks on one turn

The share of working weeks where a person keeps the same clock time all week. A week that mixes an early and a late is what people mean by "all over the place".

**What it does not tell you.** One turn all week can still be a bad week if the turn is long; the week-to-week move and the duty lengths are on pages 5 and 8.



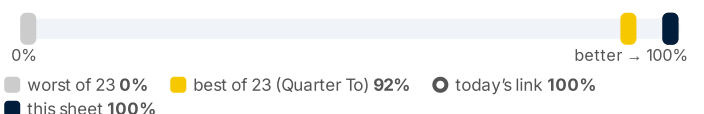
On page 5.

### IS IT FAMILIAR?

#### Shift times people work today

The share of this sheet's clock times that somebody on the 20-line link already works. It is the proxy used here for how much there is to learn and to accept.

**What it does not tell you.** Familiar is not the same as popular — a time everyone works and nobody likes still counts — and a new time in one week is not the same as one in ten. The duty table on page 5 lists the new ones.



On page 5.

## BEFORE THE FIGURES

# The decision this sheet supports

What is being decided, which criteria are hard and which are the room's, how much each figure can bear, and what no page here can tell you. Read this before the tiles.

- 1 What this is
- 2 How to read the numbers
- 3 The decision it supports (this page)
- 4 The 24-line grid, with the words explained
- 5 A week compared with today
- 6 People on duty, hour by hour
- 7 Hard limits and the December 2026 figures
- 8 The ORR fatigue factors
- 9 Beside the other proposals
- 10 The grid, ready to paste into the app

## The question

Should the CEA link for the December 2026 timetable be **Same Turns**? It is judged against today's 20-line link and the December 2026 rules, and page 9 sets it beside the other 22 proposals on the same figures.

**Same Turns** — today's link widened to 24 lines, in today's own shift times. Built by computer search: the duty table was chosen for how closely it follows the December 2026 timetable, and the 24 weeks put in the order that raises the fewest fatigue factors.

## The criteria, in three tiers

### HARD — A DESIGN MEETS THESE OR CANNOT BE RUN

- ✓ Twelve hours between duties — met, tightest 13h 30m
- ✓ Chiltern's 13-day limit — longest run 6 days
- ✓ The 35-hour contract, exactly — 35h 00m

### SOFT — THE ROOM WEIGHS THESE AGAINST EACH OTHER

- December 2026 rules met: **7 of 11** (today 3 of 11)
- Fatigue factors present: **1** (today 5)
- Demand fit weekday · Sat · Sun: **46.3** · 32.7 · 80.6 (today 51.1 · 45.7 · 65.9; lower is closer); floor **57.5** · **48.7** · **100.9** (today 68.8 · 68.7 · 72.5)
- Times worked today: **15 of 15** · full weekends off: **6** in 24 (today 4 in 20)

### PREFERENCE — STAFF'S TO STATE, NOT THE TOOL'S

- Early against late: earlies 7h 25m–8h 30m, lates 7h 30m–9h 10m
- No new clock time to learn
- The longest duty, 9h 10m, and who holds it
- Cover weeks at lines 1, 7, 13, 19

## How much to trust each figure

| FIGURE ON THESE PAGES                                                                  | WHERE IT COMES FROM                                                                                                               | WEIGHT                                      |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 0 rests under 12h · 35h 00m contract · 14/10 at the weekend                            | Computed from the cells; re-computed by the Links designer on import                                                              | <b>Firm</b>                                 |
| Longest run 6 days, inside 13                                                          | Computed; the 13-day limit's policy citation is still outstanding                                                                 | <b>Firm figure</b> , unconfirmed threshold  |
| At least four at the open, three at the close, five at 22:00, 14 and 10 at the weekend | Relayed December 2026 headcounts with no document behind them (class C)                                                           | <b>Provisional</b>                          |
| 1 fatigue factor present                                                               | The ORR's 2021 list, which it states are not limits; 4 factors still "definition to confirm"                                      | <b>Advisory only</b> — never pass or fail   |
| Demand fit 46.3 · 32.7 · 80.6; on the floor 57.5 · 48.7 · 100.9                        | Measured timetable, one share-based measure on one December 2026 timetable curve; floor takes the ticket office out of both sides | <b>Indicative</b>                           |
| 15 of 15 times are today's                                                             | Computed against the live 20-line link                                                                                            | <b>Firm</b> , but a proxy for acceptability |

## What no page here can tell you

- Whether the people who would work them accept the week shapes as written — familiarity is measured as "times worked today", which is a proxy for acceptability, not acceptability.
- Whether the roster clerk splits a cover week day-on-day-off. It is the one thing that moves the 48-hour-break figure (9 as a block, 10 split), and the link does not decide it.
- How leave and sickness cover land on a 24-line link with four cover weeks (lines 1, 7, 13, 19); nothing here models absence.
- Whether the December 2026 timetable demand curve holds on Wembley event days — the curve is a measured timetable, not a footfall count.
- Why Sunday's fit (80.6) is worse than today's (65.9) and whether anyone minds: the December 2026 headcount of ten is met by adding people, not by reshaping the day.

## Before it is frozen

Lines 6, 12, 16 hold the longest duty (9h 10m); the people who would work those weeks read them first and say in one line each what they would change. The roster office answers the cover-week question. Someone confirms the source of the 13-day limit. Only then does the factor count on page 8 mean anything, and only then is this sheet more than an aid to a conversation.

THE ROTATION

# The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

| LINE    | SUN               | MON               | TUE               | WED               | THU               | FRI               | SAT               | MON-SAT<br>hours | ALL WEEK<br>hours | DAYS WORKED<br>Mon-Sat | TURNS<br>Mon-Fri |
|---------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|-------------------|------------------------|------------------|
| 1       | SP                | SP                | SP                | SP                | SP                | SP                | SP                | 35h 00m          | 35h 00m           | —                      | —                |
| 2       | RD                | RD                | RD                | 08:00<br>16:30    | 08:00<br>16:30    | 08:00<br>16:30    | 06:20<br>14:50    | 34h 00m          | 34h 00m           | 4                      | <b>1</b>         |
| 3       | 07:15<br>15:45    | 08:00<br>16:30    | RD                | RD                | 08:00<br>16:30    | 08:00<br>16:30    | 06:20<br>14:00    | 33h 10m          | 41h 40m           | 4                      | <b>1</b>         |
| 4       | 07:15<br>15:45    | RD                | RD                | 06:20<br>13:45    | 06:20<br>13:45    | 06:20<br>13:45    | RD                | 22h 15m          | 30h 45m           | 3                      | <b>1</b>         |
| 5       | RD                | 13:30<br>22:00    | 13:30<br>22:00    | 13:30<br>22:00    | 13:30<br>22:00    | RD                | 14:30<br>22:00    | 41h 30m          | 41h 30m           | 5                      | <b>1</b>         |
| 6       | 14:30<br>23:25    | 15:15<br>23:55    | 15:15<br>23:55    | RD                | RD                | 15:15<br>23:55    | 14:45<br>23:55    | 35h 10m          | 44h 05m           | 4                      | <b>1</b>         |
| 7       | SP                | SP                | SP                | SP                | SP                | SP                | SP                | 35h 00m          | 35h 00m           | —                      | —                |
| 8       | RD                | RD                | 06:20<br>14:20    | 06:20<br>14:20    | 06:20<br>14:20    | 06:20<br>14:20    | RD                | 32h 00m          | 32h 00m           | 4                      | <b>1</b>         |
| 9       | RD                | 06:20<br>13:45    | 06:20<br>13:45    | 06:20<br>13:45    | 06:20<br>13:45    | RD                | RD                | 29h 40m          | 29h 40m           | 4                      | <b>1</b>         |
| 10      | RD                | 06:20<br>14:20    | 08:00<br>16:30    | 08:00<br>16:30    | 06:20<br>14:20    | 06:20<br>14:20    | RD                | 41h 00m          | 41h 00m           | 5                      | 2                |
| 11      | RD                | 15:15<br>23:55    | 15:15<br>23:55    | 15:15<br>23:55    | 15:15<br>23:55    | 15:15<br>23:55    | RD                | 43h 20m          | 43h 20m           | 5                      | <b>1</b>         |
| 12      | RD                | 15:15<br>23:55    | 15:15<br>23:55    | 15:15<br>23:55    | 15:15<br>23:55    | RD                | 14:45<br>23:55    | 43h 50m          | 43h 50m           | 5                      | <b>1</b>         |
| 13      | SP                | SP                | SP                | SP                | SP                | SP                | SP                | 35h 00m          | 35h 00m           | —                      | —                |
| 14      | 11:00<br>19:30    | RD                | RD                | 14:00<br>22:30    | 14:00<br>22:30    | 14:00<br>22:30    | 12:00<br>20:00    | 33h 30m          | 42h 00m           | 4                      | <b>1</b>         |
| 15      | 11:00<br>19:30    | RD                | RD                | 15:15<br>23:55    | 15:15<br>23:55    | 15:15<br>23:55    | 14:30<br>22:00    | 33h 30m          | 42h 00m           | 4                      | <b>1</b>         |
| 16      | 14:30<br>23:25    | 14:00<br>22:30    | RD                | RD                | 14:00<br>22:30    | 14:00<br>22:30    | 14:45<br>23:55    | 34h 40m          | 43h 35m           | 4                      | <b>1</b>         |
| 17      | RD                | 08:00<br>16:30    | 08:00<br>16:30    | RD                | RD                | 08:00<br>16:30    | 06:20<br>14:50    | 34h 00m          | 34h 00m           | 4                      | <b>1</b>         |
| 18      | 07:15<br>15:45    | 06:20<br>13:45    | 06:20<br>13:45    | RD                | RD                | 06:20<br>13:45    | 08:00<br>16:30    | 30h 45m          | 39h 15m           | 4                      | <b>1</b>         |
| 19      | SP                | SP                | SP                | SP                | SP                | SP                | SP                | 35h 00m          | 35h 00m           | —                      | —                |
| 20      | RD                | RD                | 13:30<br>22:00    | 13:30<br>22:00    | 13:30<br>22:00    | 13:30<br>22:00    | RD                | 34h 00m          | 34h 00m           | 4                      | <b>1</b>         |
| 21      | RD                | 08:00<br>16:30    | 08:00<br>16:30    | 08:00<br>16:30    | 08:00<br>16:30    | RD                | 06:20<br>14:50    | 42h 30m          | 42h 30m           | 5                      | <b>1</b>         |
| 22      | 07:15<br>15:45    | 06:20<br>14:20    | 06:20<br>14:20    | 06:20<br>14:20    | RD                | RD                | 08:00<br>16:30    | 32h 30m          | 41h 00m           | 4                      | <b>1</b>         |
| 23      | 14:30<br>23:25    | 13:30<br>22:00    | 14:00<br>22:30    | RD                | RD                | 13:30<br>22:00    | 14:00<br>22:30    | 34h 00m          | 42h 55m           | 4                      | 2                |
| 24      | 13:00<br>21:00    | 14:00<br>22:30    | 14:00<br>22:30    | 14:00<br>22:30    | RD                | RD                | 14:45<br>23:55    | 34h 40m          | 42h 40m           | 4                      | <b>1</b>         |
| Cover   | <b>10</b><br>SP 4 | <b>14</b><br>SP 4 | <b>14</b><br>SP 4 | <b>14</b><br>SP 4 | <b>14</b><br>SP 4 | <b>14</b><br>SP 4 | <b>14</b><br>SP 4 | 35h 00m          | 38h 34m           | 4.20                   | 18<br>one turn   |
| Minutes | 5,145             | 6,980             | 6,980             | 6,980             | 6,980             | 6,980             | 7,100             | 42,000           | 47,145            |                        |                  |

Early turns
Late turns
Cover (spare) week — four duties of seven, any turn
Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: distinct clock times Monday to Friday, **bold** where the week is one turn (one clock time Mon-Fri and one early-or-late family across every worked day — the page 1 figure) · Minutes: duty minutes per day, and Mon-Sat is the contract (20 × 35h = 42,000)

## The words on this page

**Link** — the whole rota: 24 weeks laid out as 24 lines. Everyone works line 1, then line 2, and so on round the wheel; the link needs 24 people.

**Turn** — a shift time, such as 06:20-13:45. An **early** starts before 09:00; a **late** starts after.

**Cover (spare) week** — a week with no fixed times. You work four duties of the seven days, placed by the roster clerk to cover leave and sickness.

**The contract** — 35 hours a week, averaged over the 24 lines; a cover week counts as a full contracted week.

**Line** — one week of the pattern, Sunday to Saturday. You move down a line each week, and line 24 goes back to line 1.

**Rest day (RD)** — a day off inside the contracted week. Two together make a proper break; one on its own does not count as a 48-hour break.

**Sunday** — not part of the contract. A Sunday duty is rest-day working, paid on top of the week, which is why the hours are shown Mon-Sat and all week.

**One-turn week** — the same clock time all week (shown bold in the Turns column); the easiest kind of week to live around.

# A week, and the duty table

How a working week compares with today's, and every shift time beside the ones worked today.

## The shape of a week, against today

| MEASURE                                                  | TODAY (20)          | PROPOSED (24)             |
|----------------------------------------------------------|---------------------|---------------------------|
| Working weeks that are one turn (one clock time Mon–Fri) | 7 of 16             | <b>18 of 20</b>           |
| Weeks mixing early and late turns                        | 7                   | <b>0</b>                  |
| Rest-day breaks that are two days or more                | 13 of 17            | <b>20 of 24</b>           |
| Single rest days between duties (not a 48h break)        | 4                   | <b>4</b>                  |
| Days worked in a week (lines × days)                     | 1×3, 3×4, 11×5, 1×6 | <b>6×4, 14×5</b>          |
| Cover (spare) weeks (whole weeks)                        | lines 1, 7, 12, 17  | <b>lines 1, 7, 13, 19</b> |
| Distinct shift times (all already on today's roster)     | 18                  | <b>15</b>                 |
| Mon–Sat hours, average week (the contract)               | 35h 00m             | <b>35h 00m</b>            |
| Sunday duties (not contracted; half the working lines)   | 8 of 16 lines       | <b>10 of 20 lines</b>     |

## Changed — the December 2026 headcount

| PEOPLE ON DUTY                                            | TODAY     | PROPOSED         | WHY                                                             |
|-----------------------------------------------------------|-----------|------------------|-----------------------------------------------------------------|
| Monday to Friday                                          | 11–12     | <b>14</b>        | the contract: 20 working lines × 35h has to be worked somewhere |
| Saturday                                                  | 10        | <b>14</b>        | the December 2026 figure                                        |
| Sunday                                                    | 8         | <b>10</b>        | the December 2026 figure                                        |
| Ticket office wk · Sat · Sun                              | 4 · 4 · 3 | <b>4 · 4 · 4</b> | not floor cover; Sunday one late today                          |
| Opening at 06:20 (07:15 Sunday) wk · Sat · Sun            | 4 · 4 · 3 | <b>4 · 4 · 4</b> | at least four to open, every day — met                          |
| Through to the close (23:55; 23:25 Sunday) wk · Sat · Sun | 2 · 2 · 3 | <b>3 · 4 · 3</b> | at least three to close, every day — met                        |
| Still on duty at 22:00 wk · Sat · Sun                     | 4 · 2 · 3 | <b>5 · 5 · 3</b> | at least five from 22:00, every day — not met: Sunday 3         |

## The duty table, beside today's

| TURN                                  | WK           | SAT          | SUN          | WK           | SAT          | SUN          |
|---------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| <del>06:20–13:35 7h 15m · Early</del> | <del>4</del> |              |              | <del>4</del> |              |              |
| 06:20–13:45 7h 25m · Early            | 1            |              |              | 2 +1         |              |              |
| 06:20–14:00 7h 40m · Early            |              | 1            |              |              | 1            |              |
| 06:20–14:20 8h 00m · Early            | 2            |              |              | 2            |              |              |
| 06:20–14:50 8h 30m · Early            |              | 3            |              |              | 3            |              |
| 07:15–15:45 8h 30m · Early            |              |              | 3            |              |              | 4 +1         |
| 08:00–16:30 8h 30m · Early            | 2            |              |              | 3 +1         | 2 +2         |              |
| <del>08:30–16:30 8h 00m · Early</del> |              |              | <del>4</del> |              |              | <del>4</del> |
| 11:00–19:30 8h 30m · Late             | 1            |              |              | -1           |              | 2 +2         |
| 12:00–20:00 8h 00m · Late             |              | 1            |              |              | 1            |              |
| 13:00–21:00 8h 00m · Late             |              |              | 1            |              |              | 1            |
| <del>13:30–21:00 7h 30m · Late</del>  |              | <del>4</del> |              |              | <del>4</del> |              |
| 13:30–22:00 8h 30m · Late             | 1            |              |              | 2 +1         |              |              |
| 14:00–22:30 8h 30m · Late             | 2            |              |              | 2            | 1 +1         |              |
| 14:30–22:00 7h 30m · Late             |              | 2            |              |              | 2            |              |
| 14:30–23:25 8h 55m · Late             |              |              | 3            |              |              | 3            |
| 14:45–23:55 9h 10m · Late             |              | 2            |              |              | 4 +2         |              |
| 15:15–23:55 8h 40m · Late             | 2            |              |              | 3 +1         |              |              |

Busiest weekday · Saturday · Sunday. The first three columns are what the 20-line link does now; the last three are the proposal. Green cells grew, amber shrank, a struck-through row is a time the proposal does not use.

A weekday pays 6,980 minutes and a Saturday 7,100, so five weekdays and a Saturday make exactly 20 × 35 hours = 42,000; Sunday pays 5,145, outside the contracted week. Every time is one people work today.

# People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

How to read it: each blue cell is the number of people on duty in that hour (darker blue = more people). The orange row is the timetable — how much of the day's trains that hour carries (darker orange = busier). Read the columns top to bottom: cover should be thickest where the orange is darkest. The **fit** figure at the right end of each row scores that match for the whole day.

## Monday to Friday window 06:20–23:55 · one row per distinct weekday

|                                      | 05 | 06 | 07  | 08  | 09  | 10 | 11 | 12 | 13 | 14 | 15 | 16  | 17  | 18  | 19  | 20 | 21 | 22 | 23 | FIT  |
|--------------------------------------|----|----|-----|-----|-----|----|----|----|----|----|----|-----|-----|-----|-----|----|----|----|----|------|
| TODAY MON · WED                      |    | 4  | 4   | 6   | 6   | 6  | 7  | 7  | 7  | 7  | 7  | 7   | 5   | 5   | 5   | 4  | 4  | 4  | 2  | 55.8 |
| TODAY TUE · THU · FRI                |    | 4  | 4   | 6   | 6   | 6  | 7  | 7  | 8  | 8  | 8  | 8   | 6   | 6   | 6   | 5  | 5  | 4  | 2  | 50   |
| OF WHOM TICKET OFFICE                |    | 2  | 2   | 2   | 2   | 2  | 2  | 2  | 2  | 4  | 2  | 2   | 2   | 2   | 2   | 2  | 2  | 2  |    |      |
| OF WHOM ON THE FLOOR MON · WED       |    | 2  | 2   | 4   | 4   | 4  | 5  | 5  | 5  | 3  | 5  | 5   | 3   | 3   | 3   | 2  | 2  | 2  | 2  | 83.7 |
| OF WHOM ON THE FLOOR TUE · THU · FRI |    | 2  | 2   | 4   | 4   | 4  | 5  | 5  | 6  | 4  | 6  | 6   | 4   | 4   | 4   | 3  | 3  | 2  | 2  | 64.4 |
| PROPOSED                             |    | 4  | 4   | 7   | 7   | 7  | 7  | 7  | 9  | 9  | 10 | 10  | 7   | 7   | 7   | 7  | 7  | 5  | 3  | 46.3 |
| OF WHOM TICKET OFFICE                |    | 2  | 2   | 2   | 2   | 2  | 2  | 2  | 2  | 4  | 2  | 2   | 2   | 2   | 2   | 2  | 2  | 2  |    |      |
| OF WHOM ON THE FLOOR                 |    | 2  | 2   | 5   | 5   | 5  | 5  | 5  | 7  | 5  | 8  | 8   | 5   | 5   | 5   | 5  | 5  | 3  | 3  | 57.5 |
| DEC 2026 TRAFFIC                     | 5  | 69 | 107 | 127 | 122 | 95 | 85 | 75 | 84 | 79 | 87 | 105 | 140 | 138 | 107 | 90 | 73 | 75 | 66 |      |

## Saturday window 06:20–23:55

|                       | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | FIT  |
|-----------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|
| TODAY                 |    | 4  | 4  | 4  | 4  | 4  | 4  | 5  | 6  | 9  | 6  | 6  | 6  | 6  | 6  | 5  | 4  | 2  | 2  | 45.7 |
| OF WHOM TICKET OFFICE |    | 2  | 2  | 2  | 2  | 2  | 2  | 2  | 2  | 4  | 2  | 2  | 2  | 2  | 2  | 2  | 2  |    |    |      |
| OF WHOM ON THE FLOOR  |    | 2  | 2  | 2  | 2  | 2  | 2  | 3  | 4  | 5  | 4  | 4  | 4  | 4  | 4  | 3  | 2  | 2  | 2  | 68.7 |
| PROPOSED              |    | 4  | 4  | 6  | 6  | 6  | 6  | 7  | 7  | 13 | 10 | 10 | 8  | 8  | 8  | 7  | 7  | 5  | 4  | 32.7 |
| OF WHOM TICKET OFFICE |    | 2  | 2  | 2  | 2  | 2  | 2  | 2  | 2  | 4  | 2  | 2  | 2  | 2  | 2  | 2  | 2  |    |    |      |
| OF WHOM ON THE FLOOR  |    | 2  | 2  | 4  | 4  | 4  | 4  | 5  | 5  | 9  | 8  | 8  | 6  | 6  | 6  | 5  | 5  | 5  | 4  | 48.7 |
| DEC 2026 TRAFFIC      | 6  | 24 | 58 | 75 | 81 | 92 | 88 | 77 | 64 | 68 | 72 | 84 | 82 | 67 | 64 | 79 | 78 | 62 | 41 |      |

## Sunday window 07:15–23:25

|                       | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | FIT   |
|-----------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-------|
| TODAY                 |    |    | 3  | 4  | 4  | 4  | 4  | 4  | 5  | 8  | 8  | 5  | 4  | 4  | 4  | 4  | 3  | 3  | 3  | 65.9  |
| OF WHOM TICKET OFFICE |    |    | 2  | 2  | 2  | 2  | 2  | 2  | 2  | 3  | 3  | 1  | 1  | 1  | 1  | 1  | 1  | 1  |    |       |
| OF WHOM ON THE FLOOR  |    |    | 1  | 2  | 2  | 2  | 2  | 2  | 3  | 5  | 5  | 4  | 3  | 3  | 3  | 3  | 2  | 2  | 3  | 72.5  |
| PROPOSED              |    |    | 4  | 4  | 4  | 4  | 6  | 6  | 7  | 10 | 10 | 6  | 6  | 6  | 6  | 4  | 3  | 3  | 3  | 80.6  |
| OF WHOM TICKET OFFICE |    |    | 2  | 2  | 2  | 2  | 2  | 2  | 4  | 4  | 4  | 2  | 2  | 2  | 2  | 2  | 2  | 2  |    |       |
| OF WHOM ON THE FLOOR  |    |    | 2  | 2  | 2  | 2  | 4  | 4  | 3  | 6  | 6  | 4  | 4  | 4  | 4  | 2  | 1  | 1  | 3  | 100.9 |
| DEC 2026 TRAFFIC      |    |    | 15 | 45 | 65 | 71 | 74 | 71 | 61 | 55 | 64 | 69 | 78 | 73 | 75 | 52 | 71 | 65 | 50 |       |

**Ticket office.** Not floor cover, and staffed today exactly as in the plan Monday to Saturday (two early, two late). **Sunday differs:** today two earlys and **one** late — a 14:30–23:25 closer, in the office until 22:30 — where the plan has two and two. This design does not roster the office as pairs, so its floor is everyone on duty less the plan's posts. The indented rows take it out of each side, so the floor rows compare like with like.

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. **Fit** is one number for how closely the people on duty follow the trains through the day: 0 would be a perfect match, lower is better, and today's link scores what it scores. (For the record: the squared gap between each hour's share of the day's traffic and its share of the cover, inside the day's window, measured on duty minutes — so an hour a duty only touches counts in proportion; the cells above count heads.) The fit of the average weekday is the one on page 1. Sunday: 5 December 2026 timetable movements fall after the 23:25 finish (the last at 23:54, three of them arrivals) — the standing question on whether Sunday's window moves; the window is stored per design, so the proposal can be rebuilt to either answer.

**Reading it.** The **floor** rows are the ones to read: the ticket office is staffed to its opening hours, not to the trains, so every everyone-on-duty row fits more loosely than its floor row. The fit measures **shape, not numbers** — each hour's share of the day's cover against its share of the day's trains — so an extra person in a quiet hour makes it worse even though nobody is worse off. Read it beside the heads, never alone.

# The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

**Hard limits** — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run **6** days (today: 9)  
Longest possible run is 6 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows. Worked as a BLOCK of four rather than split day-on-day-off, the answer is the same — **6** days — so this row does not depend on how a cover week is placed.  
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)  
Tightest anywhere: **13h 30m** — line 14 Fri 14:00-22:30 into Sat 12:00-20:00 (today's tightest: 12h 30m).
- ✓ **The contracted week, exactly** — **35h 00m** average Mon-Sat over the 24 lines, cover weeks counted as contracted weeks  
700h of duty a week across 20 working lines. Sundays (85.75h) sit on top as RDW, as they do today. Individual weeks range 22h 15m–43h 50m; only the average is the contract.

**The December 2026 rules** — the same rules every proposal is measured against

| RULE                                                                  | PROPOSAL                       |                                                                                                   |
|-----------------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------|
| At least four on at the open, every day                               | ✓ 4 · 4 · 4                    | 06:20; 07:15 on a Sunday                                                                          |
| At least three through to the close, every day                        | ✓ 3 · 4 · 3                    | 23:55; 23:25 on a Sunday                                                                          |
| At least five still on duty at 22:00, every day                       | × 5 · 5 · 3                    | a 22:00 finish is not "on at 22:00"                                                               |
| Fourteen on a Saturday, ten on a Sunday                               | ✓ 14 · 10                      |                                                                                                   |
| Four cover weeks, evenly spread                                       | ✓ lines 1, 7, 13, 19           | whole weeks, placed by the roster clerk                                                           |
| The ticket office: two early and two late, identical turns, every day | × ✓ Mon-Fri · ✓ Sat ·<br>× Sun | Mon-Fri 06:20–14:20 and 14:00–22:30; Sat 06:20–14:50 and 14:30–22:00; Sun from 07:15 and to 22:30 |
| Every weekday closer starts at 15:45                                  | × 15:15–23:55                  |                                                                                                   |
| At least two on the floor at every moment                             | × fewest 2 · 2 · 1             | the ticket office left out; checked every five minutes                                            |
| Handovers: 15 minutes to each closer, 20 in the ticket office         | ✓ floor ✓ · office 20 min      | on a Sunday every opener also stays 15 minutes past the last closer arrives                       |
| Sunday duties between 8h and 9h                                       | ✓ 8h00–8h55                    |                                                                                                   |
| No more shift times than today                                        | ✓ 15 against today's 18        |                                                                                                   |

## Still to settle

**4 of the 11 December 2026 rules are not met:** at least five still on duty at 22:00, every day; the ticket office: two early and two late, identical turns, every day; every weekday closer starts at 15:45; at least two on the floor at every moment. The table above gives the figures. **Sunday's finish.** Five December 2026 timetable movements fall after the 23:25 close; whether Sunday's window moves is not decided here.

## ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021).  $\triangle$  present — the pattern is in this design and worth a look, not a breach ·  $\checkmark$  clear — it is not ·  $\bullet$  standing — true of the station itself, not of any design · — does not apply here. This is the list the link is assessed against.

The Office of Rail and Road lists 25 roster patterns that tend to tire people — long runs of earlys, short gaps between duties, start times that jump about. For each one this table asks whether the pattern is in the rotation, today and proposed, and how big it is. The list is guidance: a factor present is a question for the room, never a pass or a fail, and a design showing nothing is not thereby approved.

| CODE | FACTOR                                                                                               | TODAY'S LINK                                        | PROPOSED                                             |
|------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------|
| FF1  | Night shift covering 00:00–05:00 <small>Time of day</small>                                          | --                                                  | --                                                   |
| FF2  | Early shift starting 05:00–07:00 <small>Time of day</small>                                          | $\bullet$ 24                                        | $\bullet$ 24                                         |
| FF3  | Very early shift starting before 05:00 <small>Time of day</small>                                    | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF4  | Very early shift (before 05:00) over 8h <small>Duty length</small>                                   | --                                                  | --                                                   |
| FF5  | Day shift over 12h <small>Duty length</small>                                                        | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF7  | Early shift over 10h <small>Duty length</small>                                                      | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF8b | Fewer than 2 days rest after a block of early starts <small>Recovery time</small>                    | $\triangle$ 3                                       | $\checkmark$ 0                                       |
| FF11 | More than 13 consecutive shifts without a 48h break <small>Recovery time</small>                     | $\checkmark$ 12<br>$\checkmark$ 9 as rostered       | $\checkmark$ 10<br>$\checkmark$ 9 as rostered        |
| FF10 | More than 4 consecutive 12h day shifts <small>Cumulative</small>                                     | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF15 | More than 4 consecutive early shifts in a rotating pattern <small>Cumulative</small>                 | $\triangle$ 6                                       | $\checkmark$ 4                                       |
| MRSF | More than 12 consecutive day shifts <small>Cumulative</small>                                        | $\checkmark$ 9                                      | $\checkmark$ 6                                       |
| MRSF | More than 55 hours worked in any 7-day period <small>Cumulative</small>                              | $\triangle$ 60.8                                    | $\checkmark$ 51                                      |
| MRSF | More than 7 consecutive 8h shifts (definition to confirm) <small>Cumulative</small>                  | $\triangle$ 9                                       | $\checkmark$ 6                                       |
| MRSF | More than 6 consecutive night or early shifts in a permanent pattern <small>Cumulative</small>       | --                                                  | --                                                   |
| FF17 | Backward rotating pattern (definition to confirm) <small>Circadian</small>                           | $\checkmark$ 10 backward / 12 forward               | $\checkmark$ 13 backward / 13 forward                |
| FF18 | Rotating pattern of about a week (definition to confirm) <small>Circadian</small>                    | $\bullet$ 20-line rotation · typically 4h 0m a week | $\bullet$ 24-line rotation · typically 2h 33m a week |
| FF19 | Successive start times varying by more than 2 hours (definition to confirm) <small>Circadian</small> | $\triangle$ 6                                       | $\triangle$ 1                                        |
| FF6  | Night shift over 10h <small>Night shifts</small>                                                     | --                                                  | --                                                   |
| FF8  | Less than 2 days rest after consecutive nights <small>Night shifts</small>                           | --                                                  | --                                                   |
| FF9  | Less than 14h rest in any 24h for night shifts <small>Night shifts</small>                           | --                                                  | --                                                   |
| FF12 | Only one day rest after night shifts <small>Night shifts</small>                                     | --                                                  | --                                                   |
| FF14 | More than 4 consecutive nights in a rotating pattern <small>Night shifts</small>                     | --                                                  | --                                                   |
| FF16 | More than 3 consecutive night shifts over 8h <small>Night shifts</small>                             | --                                                  | --                                                   |
| FF20 | First night shift <small>Night shifts</small>                                                        | --                                                  | --                                                   |
| FF13 | Less than 12h rest in any 24h <small>Intervals between duties</small>                                | $\checkmark$ 0                                      | $\checkmark$ 0                                       |

Present: 1 (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. **Two readings of a cover week, on the one row they move.** A cover week is worked four days of seven and the link does not say which four, so a run figure is a range. The headline number is the ceiling — the four split day-on-day-off, which supplies no 48-hour break and joins the blocks either side. It is reachable: of the 35 ways to place four duties in seven days, the 10 that leave no two rest days together produce exactly it. Worked as a BLOCK, which is what a cover week looks like on the roster, the three rest days fall together and the week always supplies a break — that is the *as rostered* figure beneath it. Checked on every row: FF11 is the only one where the two differ. Which reading applies is a question for the roster office, not for this sheet. This sheet is an aid to a conversation, not a fatigue risk assessment.

# Where it stands among the 23

Every proposal on the same figures, with today's link at the foot. This sheet's row is in bold.

**How it was made.** Built by computer search: the duty table was chosen for how closely it follows the December 2026 timetable, and the 24 weeks put in the order that raises the fewest fatigue factors. Every figure below was measured the same way for every proposal, from its grid, by the Links designer's own rule modules.

| PROPOSAL              | RULES MET   | FATIGUE FACTORS | LONGEST RUN | FULL WEEKENDS OFF | FLOOR FIT — WK · SAT · SUN |             |              | SHIFT TIMES | WORKED TODAY |
|-----------------------|-------------|-----------------|-------------|-------------------|----------------------------|-------------|--------------|-------------|--------------|
| Right Away            | 11/11       | 0               | 6           | 6                 | 20.4                       | 11.6        | 19.2         | 17          | 5            |
| Pinned Turns          | 8/11        | 0               | 6           | 6                 | 30.9                       | 31.7        | 58.1         | 16          | 9            |
| Round Times           | 8/11        | 0               | 6           | 6                 | 33.1                       | 30.8        | 58.1         | 14          | 6            |
| Quarter To            | 8/11        | 0               | 6           | 6                 | 54.9                       | 48.7        | 100.9        | 13          | 12           |
| Weekend Capped        | 7/11        | 1               | 6           | 6                 | 54.9                       | 30.9        | 65.7         | 11          | 9            |
| <b>Same Turns</b>     | <b>7/11</b> | <b>1</b>        | <b>6</b>    | <b>6</b>          | <b>57.5</b>                | <b>48.7</b> | <b>100.9</b> | <b>15</b>   | <b>15</b>    |
| By the Book           | 5/11        | 0               | 6           | 6                 | 31.9                       | 9.7         | 30.9         | 19          | 0            |
| Eight Forty           | 5/11        | 0               | 6           | 6                 | 100                        | 23.4        | 19.8         | 26          | 2            |
| Office Written In     | 4/11        | 0               | 6           | 5                 | 96.6                       | 12.7        | 69.1         | 26          | 4            |
| Gates Mended          | 4/11        | 1               | 6           | 6                 | 33.3                       | 45.4        | 39.3         | 15          | 4            |
| Light Retime          | 4/11        | 3               | 9           | 4                 | 37.2                       | 31.1        | 47           | 23          | 11           |
| Saturday Four         | 4/11        | 3               | 9           | 5                 | 37.2                       | 30.7        | 67.9         | 18          | 10           |
| Weekday Lates         | 4/11        | 5               | 9           | 6                 | 41.4                       | 19          | 67.9         | 18          | 13           |
| Fifteen Turns         | 4/11        | 7               | 9           | 2                 | 33.3                       | 47.3        | 39.3         | 15          | 5            |
| Anchored Lines        | 3/11        | 1               | 7           | 6                 | 38.1                       | 19          | 67.9         | 19          | 13           |
| Frozen Block          | 3/11        | 1               | 6           | 4                 | 38.1                       | 19          | 67.9         | 19          | 13           |
| Evening Peak          | 3/11        | 1               | 6           | 6                 | 41.9                       | 19          | 67.9         | 19          | 13           |
| Tenth Sunday          | 3/11        | 2               | 6           | 5                 | 37.2                       | 31.1        | 47           | 24          | 11           |
| Cover at Seventeen    | 3/11        | 3               | 9           | 5                 | 37.2                       | 19          | 67.9         | 19          | 13           |
| Short Closer          | 3/11        | 3               | 9           | 4                 | 37.2                       | 46.3        | 67.9         | 21          | 12           |
| Targeted Fatigue Redo | 3/11        | 4               | 9           | 5                 | 37.2                       | 19          | 67.9         | 19          | 13           |
| Three Mondays         | 3/11        | 4               | 9           | 5                 | 37.2                       | 19          | 67.9         | 19          | 13           |
| Weeks 17-18 Swapped   | 3/11        | 5               | 9           | 6                 | 37.2                       | 19          | 67.9         | 19          | 13           |
| Today's 20-line link  | 3/11        | 5               | 9           | 4                 | 68.8                       | 68.7        | 72.5         | 18          | 18           |

Ordered by rules met, then fewest fatigue factors, then the weekday floor fit — an order, not a verdict. **Rules met** counts the 11 December 2026 rules on page 7. **Floor fit** is how closely the people on the floor, the ticket office taken out, follow the trains (lower is closer; page 6). **Longest run** is days worked in a row at worst, against Chiltern's limit of 13; **full weekends off** are out of 24 (20 for today's link).

## LOAD IT

# The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

Links → Import, then paste the whole block below. Line number, then Sunday to Saturday; SP is a cover week. The Links designer re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied beside this PDF as Same-Turns-ST-24-B7-import.txt (tab-separated, pastes directly) and Same-Turns-ST-24-B7.json (the app's own format).

|    |             |             |             |             |             |             |             |
|----|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| 1  | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 2  | RD          | RD          | RD          | 08:00-16:30 | 08:00-16:30 | 08:00-16:30 | 06:20-14:50 |
| 3  | 07:15-15:45 | 08:00-16:30 | RD          | RD          | 08:00-16:30 | 08:00-16:30 | 06:20-14:00 |
| 4  | 07:15-15:45 | RD          | RD          | 06:20-13:45 | 06:20-13:45 | 06:20-13:45 | RD          |
| 5  | RD          | 13:30-22:00 | 13:30-22:00 | 13:30-22:00 | 13:30-22:00 | RD          | 14:30-22:00 |
| 6  | 14:30-23:25 | 15:15-23:55 | 15:15-23:55 | RD          | RD          | 15:15-23:55 | 14:45-23:55 |
| 7  | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 8  | RD          | RD          | 06:20-14:20 | 06:20-14:20 | 06:20-14:20 | 06:20-14:20 | RD          |
| 9  | RD          | 06:20-13:45 | 06:20-13:45 | 06:20-13:45 | 06:20-13:45 | RD          | RD          |
| 10 | RD          | 06:20-14:20 | 08:00-16:30 | 08:00-16:30 | 06:20-14:20 | 06:20-14:20 | RD          |
| 11 | RD          | 15:15-23:55 | 15:15-23:55 | 15:15-23:55 | 15:15-23:55 | 15:15-23:55 | RD          |
| 12 | RD          | 15:15-23:55 | 15:15-23:55 | 15:15-23:55 | 15:15-23:55 | RD          | 14:45-23:55 |
| 13 | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 14 | 11:00-19:30 | RD          | RD          | 14:00-22:30 | 14:00-22:30 | 14:00-22:30 | 12:00-20:00 |
| 15 | 11:00-19:30 | RD          | RD          | 15:15-23:55 | 15:15-23:55 | 15:15-23:55 | 14:30-22:00 |
| 16 | 14:30-23:25 | 14:00-22:30 | RD          | RD          | 14:00-22:30 | 14:00-22:30 | 14:45-23:55 |
| 17 | RD          | 08:00-16:30 | 08:00-16:30 | RD          | RD          | 08:00-16:30 | 06:20-14:50 |
| 18 | 07:15-15:45 | 06:20-13:45 | 06:20-13:45 | RD          | RD          | 06:20-13:45 | 08:00-16:30 |
| 19 | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 20 | RD          | RD          | 13:30-22:00 | 13:30-22:00 | 13:30-22:00 | 13:30-22:00 | RD          |
| 21 | RD          | 08:00-16:30 | 08:00-16:30 | 08:00-16:30 | 08:00-16:30 | RD          | 06:20-14:50 |
| 22 | 07:15-15:45 | 06:20-14:20 | 06:20-14:20 | 06:20-14:20 | RD          | RD          | 08:00-16:30 |
| 23 | 14:30-23:25 | 13:30-22:00 | 14:00-22:30 | RD          | RD          | 13:30-22:00 | 14:00-22:30 |
| 24 | 13:00-21:00 | 14:00-22:30 | 14:00-22:30 | 14:00-22:30 | RD          | RD          | 14:45-23:55 |

Prepared 28 September 2026. The code **d15e1b74** in every footer is the design's fingerprint — eight characters computed from these 168 cells and nothing else. If the Links designer shows the same code after you paste, you have exactly this design; if one cell differs, the code changes. Every figure in this document was computed by the Marylebone Roster app's Links modules (v24.35) from exactly these cells; import them and the Links designer's Design checks, hard limits, fatigue factors and coverage cards will restate them.